

Single point failure

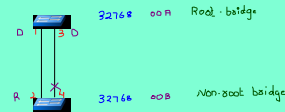
Layer 2 loop

STP

Spanning tree protocol

It is a protocol is use to

avoid layer 2 loop by using some back port



Election

Root = Root bridge

Lowest Bridge ID

BID = priority + MAC

Priority = value

Default priority = 32768

All the ports in root bridge is selected as designated port

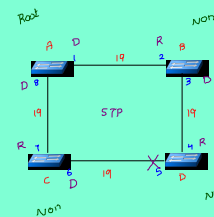
Designated port = forwarding port = open port

Now we want to find open port in non root bridge

open port = root port

Condition

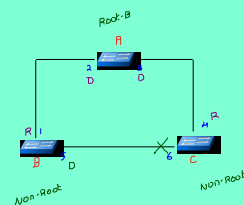
1. Lowest path cost
2. Lowest root BID
3. Lowest port id



R → D

Speed	Short cost
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2
100 Gbps	-
1 Tbps	-
10 Tbps	-

Segment
Low mac = Designated
High mac = Alternate / Backup



	Subnet mask	Total no of IP
--	-------------	----------------

/ 9		
-----	--	--

/ 30		
------	--	--

/ 28		
------	--	--

/ 18		
------	--	--

/ 10		
------	--	--